PREDICTIVE MODELING AND EXPLORATORY DATA ANALYSIS OF EMPLOYEE ATTRITION, SALARY, AND PROMOTION USING THE CTU EMPLOYEE RELATIONAL DATABASE

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Abstract

Human Resources (HR) analytics plays a critical role in understanding workforce dynamics and supporting strategic organizational decisions. This study presents a comprehensive exploratory data analysis and predictive modeling framework applied to the CTU Employee Relational Database, focusing on three key HR outcomes: employee attrition, salary prediction, and promotion to managerial roles.

Multiple relational tables were merged into a unified analytical snapshot containing approximately 300,000 employees. Feature engineering techniques were applied to derive meaningful attributes such as employee age, tenure, resignation status, and managerial status. For salary prediction, Linear Regression, Random Forest Regressor, and LightGBM models were evaluated, where LightGBM achieved the best performance with an RMSE of **13,465** and an R^2 score of **0.418**.

For attrition prediction, Logistic Regression, Random Forest, and XGBoost classifiers were compared. While Random Forest achieved the highest accuracy (**0.889**), XGBoost demonstrated superior recall for resigned employees and achieved the highest ROC-AUC score (**0.889**), making it more suitable for identifying employees at risk of leaving.

Promotion prediction was analyzed as a highly imbalanced classification problem, revealing that managerial promotions are rare and weakly correlated with standard demographic and employment features. Overall, the results highlight the effectiveness and limitations of machine learning techniques in HR analytics and provide data-driven insights for employee retention, compensation planning, and organizational development.

I. Introduction

Employee attrition, salary management, and promotion planning are among the most important challenges faced by modern organizations. High turnover rates increase recruitment and training costs, while ineffective compensation and promotion policies can negatively affect employee motivation and organizational performance.

With the growth of enterprise data systems, organizations increasingly rely on data analytics and machine learning to understand employee behavior and predict future outcomes. However, many HR analytics studies rely on simplified datasets that do not accurately represent real-world enterprise data structures.

The CTU Employee Relational Database provides a realistic HR dataset composed of multiple normalized tables with temporal information. This project aims to demonstrate how such relational data can be transformed into actionable insights by developing predictive models for:

1. employee attrition,
2. salary estimation, and
3. promotion to managerial roles.

II. Related Work

Previous research has explored employee salary and attrition prediction using machine learning techniques. Bao et al. showed that ensemble-based models such as Random Forest outperform linear models in salary prediction tasks. Yadav and Kumar applied regression techniques to institutional compensation records and demonstrated the effectiveness of feature-driven salary estimation.

Studies on attrition prediction emphasize the importance of handling class imbalance, as resignation events are typically less frequent. Boosting algorithms such as XGBoost have been shown to improve recall for minority classes.

This study extends existing work by integrating salary, attrition, and promotion analysis within a single unified dataset derived from a relational HR database.

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III. Dataset Description

A. Data Source

- Dataset: CTU Employee Relational Database
- Source: CTU Relational Learning Repository
- Format: SQL tables merged into employee_snapshot_data.parquet
- Size: ~300,000 employees, ~2.8 million salary records

B. Attributes

- The final dataset includes:
- Demographic Data: gender, birth date, hire date

- Employment Data: department name, job title, managerial status
- Compensation Data: current salary
- Engineered Features:
    - age
    - tenure
    - has_resigned
    - is_manager

C. Target Variables

- Attrition: has_resigned (binary)
- Salary: current_salary (continuous)
- Promotion: is_manager (binary)

IV. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the distribution of variables and relationships between features.

Attrition analysis shows that approximately 20% of employees have resigned, indicating moderate class imbalance. Promotion analysis reveals extreme imbalance, as managerial roles constitute less than 0.01% of employees.

Salary distribution is right-skewed with a mean salary of 69,928, indicating the presence of high-income outliers. Correlation analysis reveals:

- a positive correlation between tenure and salary (0.37),
- a negative correlation between attrition and tenure (-0.49),
- a negative correlation between attrition and age (-0.45),
- near-zero correlation between promotion and other numeric features.

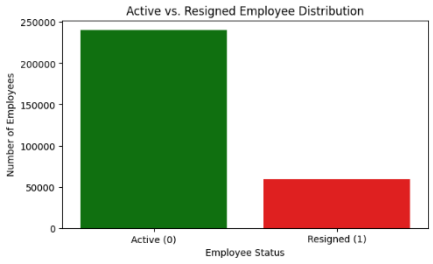


Figure 1: Active vs. Resigned Employee Distribution

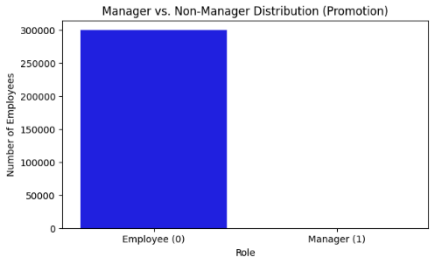


Figure 2: Manager vs. Non-Manager Distribution

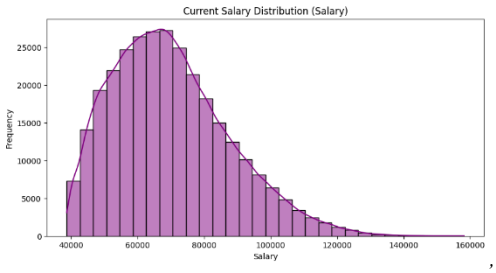


Figure 3: Salary Distribution Histogram

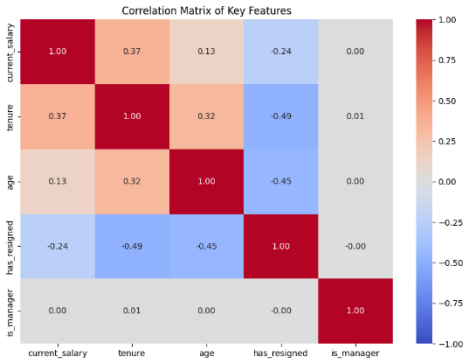


Figure 4: Correlation Matrix Heatmap

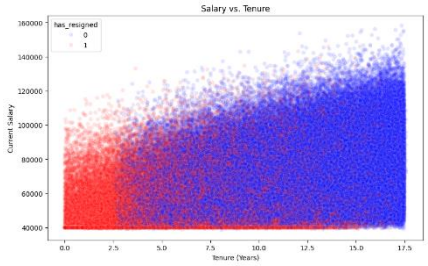


Figure 5: Salary vs. Tenure Scatter Plot

V. Methodology

A. Data Preprocessing

- Relational tables were joined using employee identifiers.
- Temporal fields were handled to distinguish active and resigned employees.
- Categorical features were encoded using One-Hot Encoding.
- Potential data leakage columns were removed.

B. Modeling Strategy

Separate models were trained for each prediction task:

- Salary Prediction: Linear Regression, Random Forest Regressor, LightGBM
- Attrition Prediction: Logistic Regression, Random Forest, XGBoost
- Promotion Prediction: Exploratory classification analysis due to extreme class imbalance

Evaluation metrics included MAE, RMSE, and R^2 for regression tasks, and Accuracy, Recall, F1-score, and ROC-AUC for classification tasks.

VI. Results

A. Salary Prediction Results

Model	MAE	RMSE	R^2
Linear Regression	10,728	13,673	0.399
Random Forest Regressor	10,605	13,536	0.411
LightGBM Regressor	**10,559**	**13,465**	**0.418**

LightGBM achieved the best overall performance.

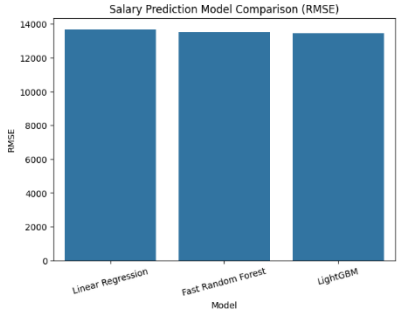


Figure 6: Salary Prediction Model Comparison (RMSE)



Figure 7: Salary Feature Importance (Random Forest)

B. Attrition Prediction Results

Model	Accuracy	Recall (Resigned)	ROC-AUC
Logistic Regression	0.872	0.50	0.87
Random Forest	**0.889**	0.53	0.88
XGBoost	0.829	**0.74**	**0.889**

XGBoost provides the best balance between recall and overall discrimination capability.

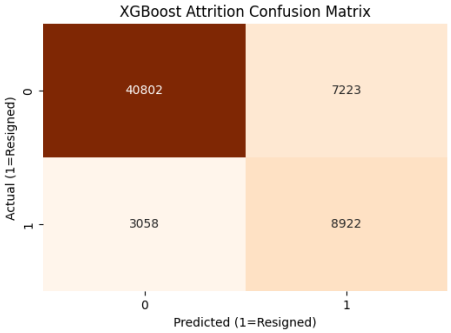


Figure 8: Confusion Matrices for Attrition Models

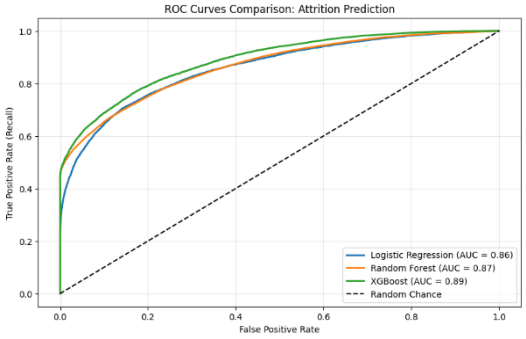


Figure 9: ROC Curve Comparison

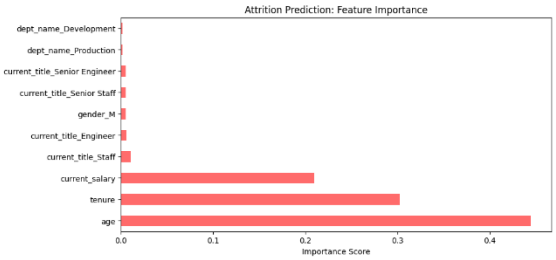


Figure 10: Attrition Feature Importance

C. Promotion Prediction Results

Promotion prediction focuses on identifying employees promoted to managerial roles, represented by the binary variable is_manager. The analysis reveals that managerial promotions are extremely rare, accounting for less than **0.01%** of all employees.

Correlation analysis indicates that promotion status has **no strong linear relationship** with age, tenure, or salary. This suggests that promotion decisions are influenced more by organizational and qualitative factors rather than measurable demographic attributes.

Due to the severe class imbalance, predictive performance is inherently limited. Tree-based models are conceptually more suitable for this task; however, the lack of positive samples restricts model generalization. As a result, promotion prediction is treated as an analytical insight rather than a fully optimized predictive task.

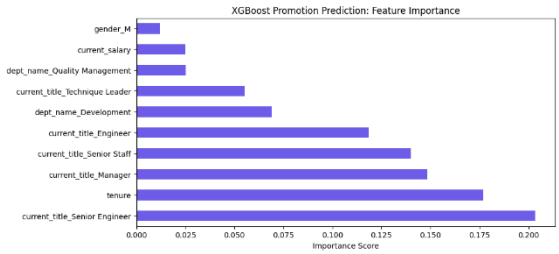


Figure 11: Promotion Feature Importance

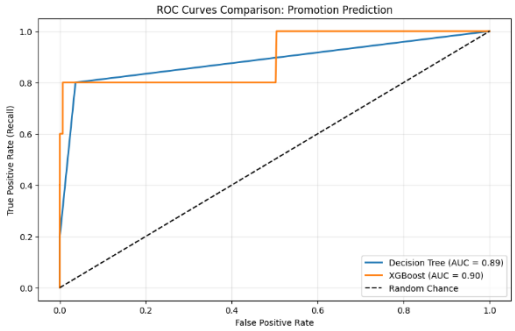


Figure 12: : ROC Curve Comparison

VII. Discussion

The results demonstrate that tenure and age are the most influential predictors for salary and attrition. Employees with longer tenure tend to earn higher salaries and are less likely to resign. Attrition prediction benefits significantly from ensemble and boosting models, particularly when recall is prioritized.

Promotion prediction highlights an important limitation of structured HR data. Unlike salary and attrition, promotions are driven by factors such as leadership ability, performance evaluations, and organizational needs, which are not captured in the dataset.

VIII. Conclusion

This study presents an end-to-end HR analytics framework using a large-scale relational employee database. By combining exploratory analysis, feature engineering, and predictive modeling, the project demonstrates how machine learning can support HR decision-making.

While salary and attrition predictions achieve strong performance, promotion prediction exposes the need for richer, qualitative data. Future work includes integrating performance metrics, deploying models in HR dashboards, and extending predictions to department-level workforce planning.

References

[1] Bao, Q. et al., “Enhancing Salary Prediction Accuracy with Advanced Machine Learning Models,” Applied and Computational Engineering, 2024.

[2] Yadav, P. K., Kumar, R., “Salary Prediction Using Regression Techniques,” SSRN, 2021.

[3] CTU Relational Learning Repository, Employee Database Dataset.